

SVKM's NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING /
SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2021-22

Programme: B.Tech Artificial Intelligence

Year: II Semester: IV

Subject: Machine Learning

Date: 12 April 2022

Marks: 100

Time: 10.00 am to 1.00 pm

Durations: 3 (hrs)

No. of Pages: 5

Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No.1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) **In all 5 questions to be attempted.**
- 4) All questions carry equal marks.
- 5) **Answer to each new question to be started on a fresh page.**
- 6) **Figures in brackets on the right hand side indicate full marks.**
- 7) **Assume Suitable data if necessary.**

Q1	Answer briefly:																											
CO-2; SO-1; BL-3	a.	A dataset contains size of TV screen and corresponding price. Simple linear regression is used to determine the relation between size and price. For the first estimated line, Sum of Squares from mean, $SSR = 800$ and Sum of Squares from line, $SSE = 200$. For the second estimated line $SSR = 780$ and $SSE = 420$. Which of the lines is the best estimation? Justify your answer.			[04]																							
CO-3; SO-1; BL-4	b.	Compare the k means and k nearest neighbour algorithms.			[04]																							
CO-3; SO-1; BL-3	c.	A classifier is used to train a machine to classify Apple, Orange and Pear. Confusion matrix for this classifier is shown below. Determine precision and recall for each class. <table border="1"><tr><td></td><td></td><td colspan="3">Predicted</td></tr><tr><td></td><td></td><td>Banana</td><td>Orange</td><td>Apple</td></tr><tr><td rowspan="3">Actual</td><td>Banana</td><td>20</td><td>3</td><td>2</td></tr><tr><td>Orange</td><td>3</td><td>25</td><td>4</td></tr><tr><td>Apple</td><td>5</td><td>6</td><td>30</td></tr></table>					Predicted					Banana	Orange	Apple	Actual	Banana	20	3	2	Orange	3	25	4	Apple	5	6	30	[04]
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		Banana	Orange	Apple																								
Actual	Banana	20	3	2																								
	Orange	3	25	4																								
	Apple	5	6	30																								
CO-1; SO-2; BL-2	d.	While there are many definitions of Machine Learning, please explain how one can define Machine Learning leveraging concepts like Tasks (T), Experience (E), and Performance (P) by taking a practical example.			[04]																							

CO-3; SO-2; BL-2	e.	Compare classification and clustering machine learning algorithms. Classify the following applications into classification and clustering machine learning algorithms. 1.. Identifying a pattern in customer shopping behaviour 2. Spam detection	[04]																																																																																																																									
Q2																																																																																																																												
CO-3; SO-2; BL-3	a.	Consider the location of the data points (x, y) as shown in the table below. Use K means algorithm to cluster the data points and show the updated cluster centers after first iteration. Assume k=2 and (1,2) and (3,6) as initial cluster centers. Use Euclidean Distance for computations. <table><tr><td>Sr. No.</td><td>x</td><td>y</td></tr><tr><td>1</td><td>1</td><td>1</td></tr><tr><td>2</td><td>1</td><td>2</td></tr><tr><td>3</td><td>4</td><td>6</td></tr><tr><td>4</td><td>3</td><td>6</td></tr><tr><td>5</td><td>4</td><td>5</td></tr><tr><td>6</td><td>2</td><td>2</td></tr><tr><td>7</td><td>2</td><td>4</td></tr></table>	Sr. No.	x	y	1	1	1	2	1	2	3	4	6	4	3	6	5	4	5	6	2	2	7	2	4	[12]																																																																																																	
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CO-3; SO-1; BL-3	b. b.	For the DBSCAN algorithm, we obtain the dissimilarity matrix as shown below for the 10 data-points. Assuming MinPts = 4, $\epsilon = 7$, determine the clusters. <table><tr><td></td><td>A1</td><td>A2</td><td>A3</td><td>A4</td><td>A5</td><td>A6</td><td>A7</td><td>A8</td><td>A9</td><td>A10</td></tr><tr><td>A1</td><td>0.0</td><td>6.1</td><td>9.2</td><td>7.7</td><td>9.3</td><td>7.3</td><td>5.1</td><td>10.2</td><td>6.1</td><td>6.0</td></tr><tr><td>A2</td><td>6.1</td><td>0.0</td><td>8.5</td><td>9.0</td><td>10.5</td><td>11.2</td><td>6.5</td><td>9.4</td><td>5.4</td><td>4.9</td></tr><tr><td>A3</td><td>9.2</td><td>8.5</td><td>0.0</td><td>5.6</td><td>6.2</td><td>3.0</td><td>8.0</td><td>2.3</td><td>8.5</td><td>7.7</td></tr><tr><td>A4</td><td>7.7</td><td>9.0</td><td>5.6</td><td>0.0</td><td>4.9</td><td>3.3</td><td>8.6</td><td>5.7</td><td>9.6</td><td>8.8</td></tr><tr><td>A5</td><td>9.3</td><td>10.5</td><td>6.2</td><td>4.9</td><td>0.0</td><td>3.8</td><td>7.5</td><td>5.9</td><td>8.9</td><td>7.6</td></tr><tr><td>A6</td><td>7.3</td><td>11.2</td><td>3.0</td><td>3.3</td><td>3.8</td><td>0.0</td><td>9.3</td><td>6.6</td><td>8.4</td><td>8.0</td></tr><tr><td>A7</td><td>5.1</td><td>6.5</td><td>8.0</td><td>8.6</td><td>7.5</td><td>9.3</td><td>0.0</td><td>8.2</td><td>3.8</td><td>4.1</td></tr><tr><td>A8</td><td>10.2</td><td>9.4</td><td>2.3</td><td>5.7</td><td>5.9</td><td>6.6</td><td>8.2</td><td>0.0</td><td>7.3</td><td>8.5</td></tr><tr><td>A9</td><td>6.1</td><td>5.4</td><td>8.5</td><td>9.6</td><td>8.9</td><td>8.4</td><td>3.8</td><td>7.3</td><td>0.0</td><td>3.3</td></tr><tr><td>A10</td><td>6.0</td><td>4.9</td><td>7.7</td><td>8.8</td><td>7.6</td><td>8.0</td><td>4.1</td><td>8.5</td><td>3.3</td><td>0.0</td></tr></table>		A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A1	0.0	6.1	9.2	7.7	9.3	7.3	5.1	10.2	6.1	6.0	A2	6.1	0.0	8.5	9.0	10.5	11.2	6.5	9.4	5.4	4.9	A3	9.2	8.5	0.0	5.6	6.2	3.0	8.0	2.3	8.5	7.7	A4	7.7	9.0	5.6	0.0	4.9	3.3	8.6	5.7	9.6	8.8	A5	9.3	10.5	6.2	4.9	0.0	3.8	7.5	5.9	8.9	7.6	A6	7.3	11.2	3.0	3.3	3.8	0.0	9.3	6.6	8.4	8.0	A7	5.1	6.5	8.0	8.6	7.5	9.3	0.0	8.2	3.8	4.1	A8	10.2	9.4	2.3	5.7	5.9	6.6	8.2	0.0	7.3	8.5	A9	6.1	5.4	8.5	9.6	8.9	8.4	3.8	7.3	0.0	3.3	A10	6.0	4.9	7.7	8.8	7.6	8.0	4.1	8.5	3.3	0.0	[08]
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Q3																																							
CO-2; SO-2; BL-3	a.	Consider the following data for height vs weight for new born babies. <table><tr><td>x (age in months)</td><td>y (weight in kg)</td></tr><tr><td>0</td><td>2.5</td></tr><tr><td>1</td><td>3.4</td></tr><tr><td>2</td><td>4.3</td></tr><tr><td>3</td><td>5</td></tr></table> Fit a simple linear regression model on this data. Calculate the following: a) Slope of the line (b_1) b) Intercept (b_0) c) Write the equation for the regression line. d) Determine the predicted annual increment for an employee who has 3.5 years of experience. Based on the regression line, calculate the following: e) SSR f) SSE g) SSTO h) R^2	x (age in months)	y (weight in kg)	0	2.5	1	3.4	2	4.3	3	5	[12]																										
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CO-1; SO-1; BL-4	b.	Justify the advantage of using the ensemble methods in machine learning at the cost of increased complexity with the help of an example.	[08]																																				
Q4																																							
CO-3; SO-1; BL-4	a.	Consider the dissimilarity matrix for the data points. Perform agglomerative clustering on the data points and draw the dendrogram. Use Euclidean distance as a distance measure. <table><tr><td></td><td>P</td><td>Q</td><td>R</td><td>S</td><td>T</td></tr><tr><td>P</td><td>0</td><td></td><td></td><td></td><td></td></tr><tr><td>Q</td><td>7</td><td>0</td><td></td><td></td><td></td></tr><tr><td>R</td><td>2</td><td>4</td><td>0</td><td></td><td></td></tr><tr><td>S</td><td>3</td><td>8</td><td>8</td><td>0</td><td></td></tr><tr><td>T</td><td>5</td><td>10</td><td>6</td><td>2</td><td>0</td></tr></table>		P	Q	R	S	T	P	0					Q	7	0				R	2	4	0			S	3	8	8	0		T	5	10	6	2	0	[12]
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CO-3; SO-2; BL-4	b.	Bring out the advantage of Adaboost over random forest.	[08]																																				

Q5																																																																																			
CO-3; SO-2; BL-3	a.	What is a Decision tree method / classifier and what are the most important components of the tree structure? Consider the following dataset. Decision Tree classifier based on entropy is used to classify whether the LED TV is faulty or not. Determine which attribute should be used as the root node. If entropy at any of the nodes is 0 then does this node require further spilt? Justify your answer. <table><tr><th>Sr. No.</th><th>Picture Quality</th><th>Sound Quality</th><th>Age</th><th>Class</th></tr><tr><td>1</td><td>Poor</td><td>Good</td><td>2</td><td>Faulty</td></tr><tr><td>2</td><td>Poor</td><td>Excellent</td><td>1</td><td>Faulty</td></tr><tr><td>3</td><td>Good</td><td>Poor</td><td>2</td><td>Faulty</td></tr><tr><td>4</td><td>Good</td><td>Poor</td><td>2</td><td>Faulty</td></tr><tr><td>5</td><td>Good</td><td>Excellent</td><td>1</td><td>Not Faulty</td></tr><tr><td>6</td><td>Good</td><td>Good</td><td>1</td><td>Not Faulty</td></tr><tr><td>7</td><td>Good</td><td>Good</td><td>2</td><td>Faulty</td></tr><tr><td>8</td><td>Excellent</td><td>Good</td><td>1</td><td>Faulty</td></tr><tr><td>9</td><td>Excellent</td><td>Excellent</td><td>1</td><td>Not Faulty</td></tr><tr><td>10</td><td>Excellent</td><td>Good</td><td>2</td><td>Not Faulty</td></tr><tr><td>11</td><td>Good</td><td>Good</td><td>2</td><td>Faulty</td></tr><tr><td>12</td><td>Good</td><td>Good</td><td>2</td><td>Faulty</td></tr><tr><td>13</td><td>Good</td><td>Good</td><td>1</td><td>Not Faulty</td></tr><tr><td>14</td><td>Excellent</td><td>Excellent</td><td>1</td><td>Not Faulty</td></tr><tr><td>15</td><td>Excellent</td><td>Good</td><td>1</td><td>Not Faulty</td></tr></table>	Sr. No.	Picture Quality	Sound Quality	Age	Class	1	Poor	Good	2	Faulty	2	Poor	Excellent	1	Faulty	3	Good	Poor	2	Faulty	4	Good	Poor	2	Faulty	5	Good	Excellent	1	Not Faulty	6	Good	Good	1	Not Faulty	7	Good	Good	2	Faulty	8	Excellent	Good	1	Faulty	9	Excellent	Excellent	1	Not Faulty	10	Excellent	Good	2	Not Faulty	11	Good	Good	2	Faulty	12	Good	Good	2	Faulty	13	Good	Good	1	Not Faulty	14	Excellent	Excellent	1	Not Faulty	15	Excellent	Good	1	Not Faulty	[12]
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CO-2; SO-2; BL-3	b.	Using an example show how F-distribution is used to accept or reject null hypothesis for slope. Assume that confidence interval is 80%.	[08]																																																																																
Q6																																																																																			
CO-3; SO-2; BL-3	a.	Consider the following dataset. Naïve Bayes Classifier is used to determine whether the person cheats. Predict whether the person with Tid 11 will cheat or not. <table><tr><th>Tid</th><th>Refund</th><th>Marital status</th><th>Taxable income</th><th>Cheat</th></tr><tr><td>1</td><td>Yes</td><td>Single</td><td>125K</td><td>No</td></tr><tr><td>2</td><td>No</td><td>Married</td><td>100K</td><td>No</td></tr><tr><td>3</td><td>No</td><td>Single</td><td>70K</td><td>No</td></tr><tr><td>4</td><td>Yes</td><td>Married</td><td>120K</td><td>No</td></tr><tr><td>5</td><td>No</td><td>Divorced</td><td>95K</td><td>Yes</td></tr><tr><td>6</td><td>No</td><td>Married</td><td>60K</td><td>No</td></tr><tr><td>7</td><td>Yes</td><td>Divorced</td><td>220K</td><td>No</td></tr><tr><td>8</td><td>No</td><td>Single</td><td>85K</td><td>Yes</td></tr><tr><td>9</td><td>No</td><td>Married</td><td>75K</td><td>No</td></tr><tr><td>10</td><td>No</td><td>Single</td><td>90K</td><td>Yes</td></tr><tr><td>11</td><td>Yes</td><td>Divorced</td><td>90K</td><td>??</td></tr></table>	Tid	Refund	Marital status	Taxable income	Cheat	1	Yes	Single	125K	No	2	No	Married	100K	No	3	No	Single	70K	No	4	Yes	Married	120K	No	5	No	Divorced	95K	Yes	6	No	Married	60K	No	7	Yes	Divorced	220K	No	8	No	Single	85K	Yes	9	No	Married	75K	No	10	No	Single	90K	Yes	11	Yes	Divorced	90K	??	[12]																				
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CO-3; SO-2; BL-4	b.	Consider the sample dataset shown below. Design a SVM classifier using two support vectors.	[08]																											
		<table><tr><th>Feature 1</th><th>Feature 2</th><th>Class</th></tr><tr><td>1</td><td>1</td><td>A</td></tr><tr><td>2</td><td>1</td><td>A</td></tr><tr><td>3</td><td>1</td><td>A</td></tr><tr><td>2</td><td>2</td><td>A</td></tr><tr><td>-1</td><td>-1</td><td>B</td></tr><tr><td>-2</td><td>-1</td><td>B</td></tr><tr><td>-3</td><td>-1</td><td>B</td></tr><tr><td>-2</td><td>-2</td><td>B</td></tr></table>	Feature 1	Feature 2	Class	1	1	A	2	1	A	3	1	A	2	2	A	-1	-1	B	-2	-1	B	-3	-1	B	-2	-2	B	
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